Traffic Simulator

User Manual

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1 Introduction

1.1 About

This document gives an in-depth explanation of how to use the software Traffic Simulator. The software is written in Java 17 and was developed as a project for the course System- och programvaruutveckling (DT117G) at Örebro University.

1.2 Overview

Traffic Simulator is a software for simulating traffic scenarios, of which there are several to choose from (scenarios are called maps in Traffic Simulator). The simulator features a visual representation of the simulation, to let the user review things like traffic flow in "real time", that is, as the simulation runs.

The simulation can be sped up/down, paused, stopped or reset at any time. Additionally, Traffic Simulator has settings for configuring different simulation parameters before the simulation starts. These include settings for traffic inflow, the preferred speed of vehicles and for some maps, the length of the road.

Useful statistics are shown during the simulation. These are updated in "real time" and include the current number of vehicles in the simulation, average speed and inflow/outflow of vehicles. When the simulation is stopped, a statistics summary is presented, with the option to export to a CSV file for further processing.

There is also an option to run simulations without visuals. Instead, the statistics summary is shown as soon as the host computer has finished the simulation. This can be useful to run multiple simulations quickly and compare their statistics after the fact.

2 Simulation

This section describes the everything related to the simulation mode, from the inner workings of the simulation engine to its visual represantion and available user controls.

2.1 Simulation Engine

Traffic Simulator uses a simulation model based on the Nagel-Schreckenberg model. This uses discrete time, and hence the simulation moves in **steps** or **ticks**.

Roads are divided into discrete cells, and every car in the simulation moves zero or more cells each tick. Each car has a preferred speed, and tries to move accordingly. The preferred speed is randomised at spawn time for each car between the min and max values as set in the settings menu.

To make the simulator behave like traffic, each car looks ahead and adjusts its speed accordingly: if no obstacle (including other vehicles) is seen, the car tries to keep its preferred speed, otherwise it slows down or stops to avoid collisions.

The scale of the model translates to the real world as follows:

- One cell represents 7.5 m (meters), as proposed by the Nagel-Schreckenberg model.
- One tick represents 2.7 s (seconds).
- A speed of 1 cell/tick then corresponds to 10 km/h in the real world.

2.2 Simulation Visuals

While the visual elements of Traffic Simulator are (hopefully) rather self explanatory, here is a rundown of the basics.



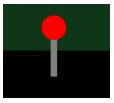
Piece of road



Intersection



Car on a road



Traffic light

2.3 Maps

Maps are models of particular traffic scenarios. The following maps are implemented:



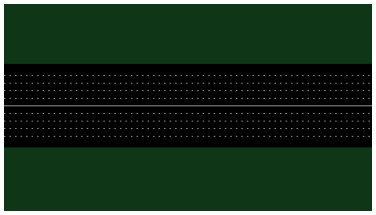
One way street: A simple one way street with a traffic light.



T Intersection: A one way street with a connecting one way street in the middle.



Highway: A highway with two lanes in each direction. Overtakes are implemented, to allow faster driving vehicles to pass slower driving ones.



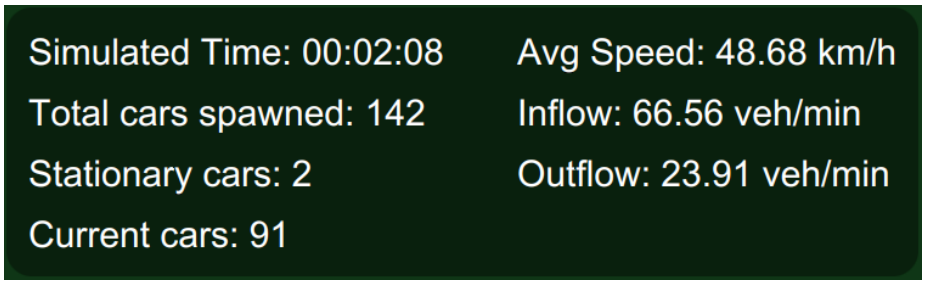
Autobahn: A large highway with five lanes in each direction. Supports overtakes in all lanes, and outer lanes has an extra speed up, to ensure more realistic behavior and keep the overtaking car from getting stuck in the outer lane.



Grid of roads: A network of roads and intersecions. Available in two versions, where the second one has traffic lights at two of the intersections.

2.4 Live Statistics

A running simulation shows live statistics. Below is a brief explanation of what these include.



Simulated time from when the simulation started. *(27x faster than actual time at 100% time scaling).*

Total cars spawned during the simulation.

The number of currently **stationary cars**.

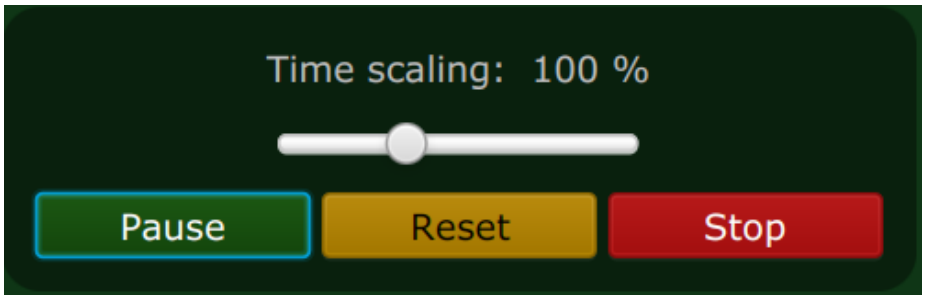
The number of **current cars** in the simulation.

Current **average speed**.

Current **inflow** and **outflow** of cars, respectively. *Shown in vehicles per simulated minute.*

2.5 Playback Controls

These are the controls available when running a simulation with visuals.



Top to bottom, left to right:

1. **Time scaling** can be adjusted while the simulation is running. This will speed up/slow down the simulation; that is, the simulated time will run fast or slower. *Tip: pressing + or - on the computer's keyboard will increment/decrement this control!*

2. **Pause**/**Resume** the simulation (and the visuals). *Tip: The simulation can also be paused using the space bar key!*

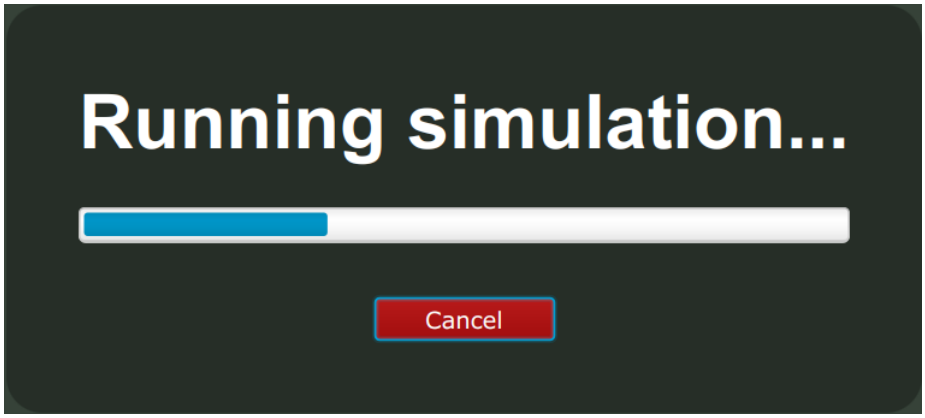
3. **Reset** simulation. This will abort the running simulation and start a new one with the same settings (even if paused). *Tip: The simulation can also be reset using the backspace key!*

4. **Stop** the simulation and show statistics summary. *Tip: The simulation can also be stopped using the escape key!*

2.6 Offline Simulation

Traffic Simulator allows for running simulations without any visuals. This is much faster than running simulations in "real time", and can be useful for running very long simulations or multiple simulations quickly. The statistics summary is still shown and the results can be exported to a CSV file.

When running a simulation in offline mode, the graphical interface shows a progress bar and a button for cancelling the simulation.



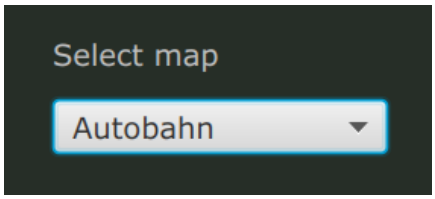
3 Settings Menu

Traffic simulator allows for a number of user settings before starting a simulation. How to use them is described below.

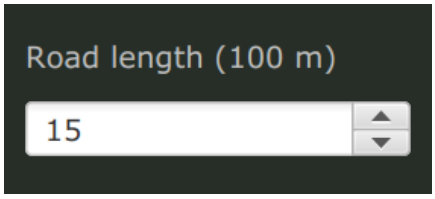
Besides starting a new simulation, the menu lets the user exit the program or quickly open this document in their systems default PDF reader.

3.1 Settings

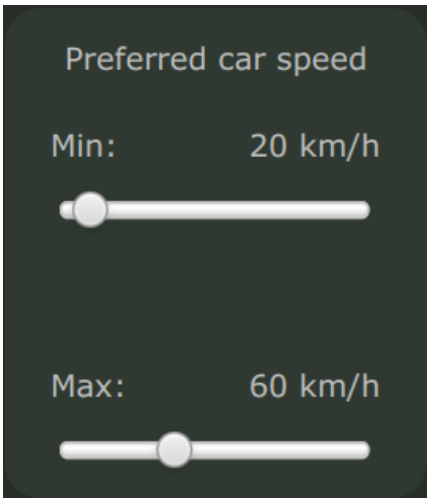
Here follows a list of the avabilable settings in the settings menu.



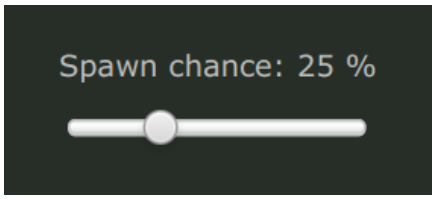
The **map selector** is used to select the map, or scenario, to run a simulation on (see section 2.3 on page 3).



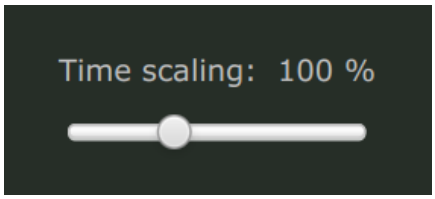
Road length decides the number of cells in the main road. *Only applicable to certain maps.* Also note the visual rendering of vehicles may appear very small for large values.



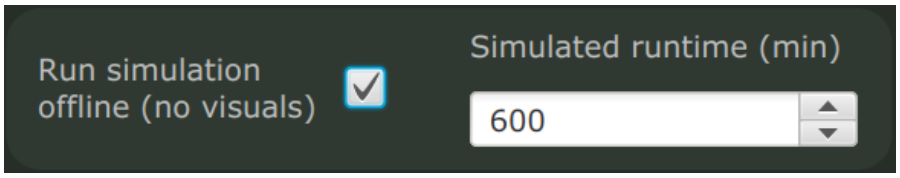
The **preferred car speed** is randomised when a new car is spawned. This settings allows for adjusting the range within which the randomisation happens, that is, the minimum and maximum preferred speed for simulated cars.



Spawn chance decides how likely it is that a new car spawns each tick.



Set the initial **time scaling** for the simulation. Can be adjusted while the simulation is running. *Not active for offline mode.*



The **run simulation offline** toggle allows for running simulations without visuals, and get the statistics summary at the end. Useful for large simulations since it greatly speeds up the simulation time.

Simulated runtime is only active when running offline simulations. This decides for how long the simulation will run, in simulated minutes.

3.2 Navigation Buttons



From the left:

Start a new simulation with the current settings.

Help opens this document in your default PDF viewer.

Exit program to desktop.

4 Statistics Page

When a simulation is stopped (or completed/cancelled if run in offline mode), a summary of the simulation statistics is shown.

4.1 Statistics Summary

The statistics summary shows a summary of the simulation and is similar to the live statistics discussed in section 2.4 (page 4), but with a few differences:

```
--- FINAL REPORT ---          --- SETTINGS ---
Simulated Time: 74:01:28      Map: Autobahn
Total Cars Spawned: 330384    Preferred Car Speed: 40-60 km/h
Total Cars Exited: 330289     Spawn Probability: 25%/tick
Average Speed: 53.58 km/h     Road Length: 1500 m
InFlow: 74.39 veh/min
OutFlow: 74.36 veh/min
Peak traffic: 131 cars        Runtime: 00:00:04
```

Total cars exited is shown, and refers to the number of cars that have exited the simulation by driving out of the simulated area.

Average speed, **InFlow** and **InFlow** now shows the average for the whole simulation.

Peak traffic is the maximum number of current cars (cars simultaneously inside the simulation) at any one time during the simulation.

Runtime is the total (real world) time for which the simulation was running. Unlike simulated time, this is not affected by time scaling and is also shown after running a simulation in offline mode.

Also, the **settings** used for the simulation are shown.

4.2 Navigation Buttons

From the statistics summary, the user can either exit the program or return the the menu for running another simulation. There is also an option to export the statistcs summary to a CSV file.



1. **Back to Menu** shows the settings menu, where a new simulation can be started.
2. **Export to CSV** lets the simulation statistics summary be exported to a CSV file, for further processing or importing to a spreadsheet.
3. **Exit** program to desktop.

4.3 Export Statistics to CSV

A useful feature of Traffic Simulator is the option to export the statistics summary of a simulation after it has run. This can then be imported into a spreadsheet program or data analysis software for comparing multiple simulations with slightly different settings, or any kind of further processing.

When pressing the button to export to a CSV file, a file dialog is shown, where a location and file name for the CSV file can be selected, as shown below.

